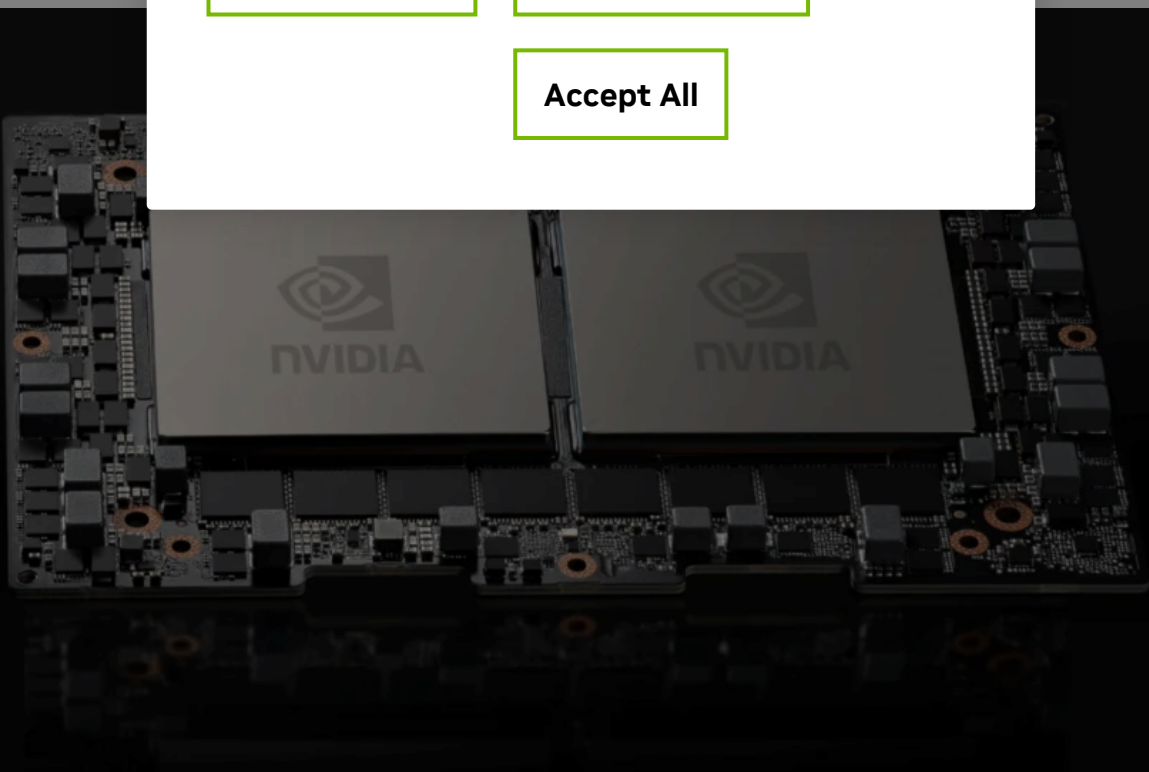




NVIDIA High Bandwidth Efficient Center

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Dec 05, 2025

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By [Praveen Menon](#) and [Ivan Goldwasser](#)

Since its debut in 2023, the NVIDIA Grace CPU has experienced rapid adoption across data centers, setting new benchmarks for performance and efficiency in high-performance workloads.

Grace integrates Armv9-A cores, 114 MB unified L3 cache, 480 GB of LPDDR5X, and a breakthrough bandwidth of 900 GB/s NVLink-C2C.

In this blog post, we explore the architecture of the Grace CPU, its benefits for memory access (NUMA) monolithic design, and how it addresses the challenges of scaling AI workloads. Us.

Single N

Grace CPU features a unified architecture that integrates Armv9-A cores, 114 MB unified L3 cache, 480 GB of LPDDR5X, and a breakthrough bandwidth of 900 GB/s NVLink-C2C. This design ensures consistent performance across all cores, eliminating cross-NUMA transfers and the associated performance penalties.

Grace's unified cache and memory subsystem provides optimal inter-core latencies and higher cache hit rates without the die-to-die hops inherent to multi-chip design.

In cloud environments where smaller-sized virtual machines (VMs) are more prevalent among end users, Grace's unified architecture enables each VM to access the full memory subsystem—an advantage over chiplet-based designs that partition memory across multiple dies. Legacy multi-chip architectures often require meticulous core pinning to maintain performance consistency, and when under-utilized, distributing workloads can incur additional power overhead from activating multiple chiplets.

Figure 1 below shows the NVIDIA SCF, which serves as the backbone, enabling the 72 Arm Neoverse cores, 114 MB unified L3 cache, 480 GB of LPDDR5X, and the 900 GB/s NVLink-C2C, all functioning as one coherent system on a single monolithic die. The illumination

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Figure 1. NVIDIA SCF enabling Neoverse cores & the memory subsystem in a monolithic die

Optimal memory bandwidth scaling with core count

NVIDIA Grace CPU offers an optimal balance of core count and memory bandwidth, connected by a vast unified mesh that moves data across the system. Many data analytics, Extract, Transform, Load ([ETL](#)), and HPC workloads move vast amounts of data between cores, caches, and memory, demanding a well-balanced ratio of memory bandwidth to total core count.

As shown in Figure 2, STREAM benchmark measurements highlight Grace's achievable memory bandwidth advantage. This benchmark is specifically designed to overwhelm CPU

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Figure 2. The Grace C70 accelerates AI workloads up to 10x faster than x86 servers

Grace surpasses ARM and x86 in bandwidth efficiency. The strength lies in bandwidth efficiency. Grace delivers higher bandwidth performance than even computing. Grace provides up to 1.8x higher per-core memory bandwidth compared to competing SoCs, enabling every core to efficiently process massive data flows simultaneously across the mesh fabric and memory subsystem.

Grace for big data workloads

The benefits of Grace's unified fabric and higher per-core memory bandwidth are evident in real-world data analytics workloads. For instance, the PageRank graph algorithm, part of the Graph Algorithm Platform Benchmark Suite (GAPBS), is a key workload for assessing system performance in graph analytics and big data. It's designed to iteratively calculate a score of importance for every node in a massive graph, simulating real-world processes like ranking websites or analyzing social networks. The PageRank benchmark stresses both inter-core communication and the ability to efficiently handle large random, scattered memory access patterns.

second (TEPS), scales consistently with core count on the Grace CPU. This is made possible by the unified SCF, which enables seamless data movement across cores, caches, and the memory subsystem.

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Figure 3. NVIDIA Grace CPU performance vs. x86 CPUs in data analytics and HPC workloads

In contrast, chiplet-based x86 designs have uneven and limited scaling due to fragmented mesh and inter-die latencies. While fine-grained core pinning can yield modest gains on chiplet architectures, it's often counterproductive for data analytics workloads and adds engineering overhead in real-world deployments. This problem is further exacerbated in cloud computing with smaller-sized VMs, where distributing cores may not be an option.

Grace outperforms x86 CPUs across data analytics and HPC workloads

Grace is the first server CPU with high-performance LPDDR5X memory and a fully coherent CPU-GPU interconnect (NVLink-C2C) running at 900 GB/s. This enables the next generation of AI factory and big data workloads.

Figure 4 shows the performance-per-watt comparison of Grace vs AMD Epyc Turin in a two-socketed system. NVIDIA Grace offers up to 3x higher performance per watt across key

ownership.

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Figure 4. NVIDIA Gr

dual-socket

Grace delivers high performance at low power

A key advantage of the Grace CPU's power-efficient architecture is its ability to maintain high performance even when operating under lower power caps, which limit CPU-module power to reduce total rack energy and cooling demands. As data centers push towards higher compute density and tighter energy budgets, efficiency becomes increasingly important. Figure 5 shows Grace's relative performance at several power-cap levels compared to its 250W baseline across identical workloads. Grace sustains over 90% performance at 200W and ~80% at 150W, enabling significant energy savings with minimal performance trade-offs.

This capability enables operators to tune for efficiency without sacrificing meaningful compute performance, maximizing rack-level density and reducing cooling costs in power-constrained environments.

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Fig

Grace achieves this design with high-performance SoC die that minimize movement overhead. It runs faster at lower power. It maintains performance headroom even under constrained power, making it ideal for modern, power-constrained environments spanning hyperscale deployments, high-performance edge, storage, CDN, high-performance computing (HPC), and other power-constrained use cases.

Powering the next generation of data factories

Grace sets the foundation of the NVIDIA CPU roadmap, delivering leadership performance, power efficiency, consistent core scaling, optimal mesh, and memory bandwidth in a single NUMA design, ideally suited for a diverse range of server deployments. Looking forward, we are very excited for the upcoming launch of our next-generation server CPU Vera, with 88 custom Arm Cores with multi-threading, a larger mesh, 1.2TB/s memory bandwidth, and 1.8 TB/s NVlink-C2C to drive the next generation of data compute.

NVIDIA Grace Superchip 480GB
OS: Ubuntu 24.04 LTS Compilers
Data Analytics: HiBench+K-means
arXiv:1508.03619 [cs.DC], 2015
Database: Clickhouse Phoronix
HPC: CFD: OpenFOAM v2406 W
Compression: Snappy (Commit
Microservices: Google Protobufs
Data Analytics: HiBench+K-means
NVIDIA Grace CPU C1 OS: Ubuntu

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Praveen is a senior technical marketing engineer focused on accelerated computing platforms in the data center at NVIDIA. Previously, Praveen held various roles across marketing, performance, and silicon engineering. He holds a master's degree in electrical and computer engineering from the University of Arizona

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Ivan leads product marketing for the Data Center CPU products for NVIDIA. Previously, Ivan worked in various marketing and strategy roles in the technology sector. Ivan has an MBA from Georgetown's McDonough School of Business and a bachelor's degree in chemical engineering from Texas A&M University.

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NVIDIA Grace CPU Integrates with the Arm Software Ecosystem

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**Advancing AI
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Grace

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Revolutionizing Data Center Efficiency with the NVIDIA Grace Family

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